

Accuracy assessment of MODIS/Terra snow cover product for parts of Indian Himalayas

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Snow cover mapping is important for snow and glacier-related research. The spatial and temporal distribution of snow cover area is a fundamental input to the atmospheric models, snowmelt runoff models and climate models, as well as other applications. Daily snow cover maps from Moderate Resolution Imaging Spectroradiometer (MODIS) Terra satellite were retrieved for the period between 2004 and 2007, and pixels in these images were classified as cloud, snow or snow-free. These images have then been compared with ground snow depth (SD) measurements from the four observatories located at different parts of Himalayas. Comparison of snow maps with *in situ* data showed good agreement with overall accuracies in between 78.15 and 95.60%. When snow cover was less, MODIS data were found to be less accurate in mapping snow cover region. As the SD increases, the accuracy of MODIS snow cover maps also increases.

Keywords: remote sensing; snow cover; MODIS; Himalayas; snow hydrology

Introduction

More than 80% of the total fresh water is in the form of frozen state on the earth (Dozier 1989). About 30% of the earth's land surface is seasonally covered by snow and contributes to the largest proportion of river discharge over major portions of middle and high altitudes. Snowmelt accounts for 50–80% of the annual stream flow in many areas of the world, including the Sierra Neveada, the Rockies, the Alps, the Andes and Himalayan Mountains (Ferris & Congalton 1989). In India, Glaciology study focuses on water resources, climate and glacial morphology with special attention to the three balances that is mass balance, water balance and energy balance (Vohra 1999). Snow cover plays an important role in snow hydrology, climatology, ecology and snow-related disaster studies like flood outburst and avalanche.

The seasonal snow line descends to an altitude of 2800 m in the eastern part and 2000 m in the western part of the Himalayas by February. Annual snowfall in the Himalayas varies from as low as 93 mm to as high as 3400 mm. As the snowmelt commences by the month of March, the snow line starts receding upward and by the middle of June it moves up to an altitude of 4500 m. (Sud & Saxena 1999).

In order to reduce damages and economic loss, a high accuracy of runoff simulation is essential. Improvements of snowmelt runoff models are possible through accurate

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determination of spatial input parameters like rainfall, snow distribution, snow properties, soil moisture, etc. Monitoring and modelling of these parameters are difficult and often missing or are of low quality at higher altitudes of Indian Himalayas due to limited amount of ground-based data. Remote-sensing data are an attractive alternative to ground-based data particularly to obtain snow information. Snow cover area monitoring by remote sensing gained importance when it started being utilized in the snowmelt run-off model since 1975 (Martinec 1975). Snow cover can be detected and monitored by using different remote sensors from high-resolution IRS and Landsat to medium resolution MODIS, SPOT and NOAA-AVHRR. Selection of the sensor depends upon the spatial and temporal resolution requirement of the application.

The MODIS snow cover products (Hall et al. 2002; Hall & Riggs 2007) have been evaluated in various places like Canada (Simic et al. 2004), Austria (Parajika & Blosch 2006), the USA (Ault et al. 2006), Turkey (Sorman et al. 2007), (Akyurek et al. 2007) India (Negi et al. 2008), China (Wang et al. 2008; Xiaodong et al. 2011) and also Global level (Salomonson & Appel 2004) etc. The accuracy of MODIS snow cover maps varies from as low as 21% to as high as 100% with an absolute accuracy of around 93% based on the cloud coverage, snow depth (SD), land-cover type, snow condition and season (Klein & Barnett 2003; Tekeli et al. 2005; Parajika & Blosch 2006; Flueraru et al. 2006; Hall & Riggs 2007; Liang et al. 2008).

MODIS snow cover products have been used for different purposes like hydrology (Rango et al. 2004; Brown et al. 2008; Bavera & Michele 2009), climatology (Nonaka et al. 2008), ecology (Vikhamar & Solberg 2003) and avalanche studies (Negi et al. 2007). These images are very useful in real-time modelling because the complete processing time including generation of end-product and dissemination to the users takes approximately 2 h after reception of real-time imagery (Negi et al. 2007). The fully automated nature of the MODIS snow cover algorithms also make the products useful for developing and augmenting long-term snow cover records and climate-data records (Hall et al. 2010).

The areal distribution of seasonal snow cover needs to be mapped onto meaningful temporal and spatial scales in the Indian Himalayas to understand how the snow cover distribution varies under various climatic change scenarios. MODIS snow cover maps are appropriate due to their high-temporal resolution, low cost, easy access and relatively high spatial resolution. Accuracy assessment of MODIS snow cover maps are useful for hydrological, ecological and climate change studies in the Indian Himalayas.

Study area, data and methodology

Purpose

The National Snow and Ice Data Centre (NSIDC) provides a valuable service of distributing snow cover maps to the public, free-of-charge and on a global scale. The aim of the present study is to validate the performance of MOD10A1 snow cover product of MODIS, distributed by the NSIDC for parts of the Indian Himalayas, with corresponding observatory data obtained from Snow and Avalanche Study Establishment (SASE). Analysis of data has been done for the period starting from October 2004 to September 2007.

Study area

The Himalayas situated in the northern part of the Indian subcontinent are approximately 2500 km long with an average altitude of about 6000 m (National Institute of

Hydrology 2001). The Himalayas are the world's highest mountainous region, and have the highest peaks including the Mount Everest (8848 m) and K2 (8611 m). The Himalayas have over 300 glaciers exceeding the altitude of 6000 m (Singh et al. 2008). The 73 km long Siachen Glacier, situated in the north-west part of the Indian Himalayan snow and glacier-covered region, is the longest glacier in the entire Indian Himalayan ranges and second longest glacier outside the polar region (Singh et al. 2008).

Many mighty rivers originate from the permanent snow fields at higher altitudes. Ganga, Indus and Brahmaputra are the major basins for north India originating from the Himalayas. The total glaciated area in these three river basins put together is 78,040 km² (Space Application Centre 2010), having highly variable terrain and meteorological conditions. In the Himalayas, massive snow accumulation leads to disasters like glacier lake outburst floods (Hewitt 1982; Yamada 1993; Richardson & Reynolds 2000) and landslides (Ganju et al. 2002) which cause loss of lives and disrupts the infrastructure. It is essential to understand these complicated basins and is best possible by the remote-sensing method

SASE observatories

There are a few observatories maintained by SASE under the Ministry of Defence, India. Data have been collected from four observatories namely Bhojwasa (30°55' 0.036" N, 79°03'03.778" E), Patsio (32°45'17.89"N, 77°15'43.13" E), Solang (32°19'04" N, 77°09'21" E) and Dhundi (32°21'19.2" N, 77°07'35.67" E). Figure 1 shows the spatial distribution of these four meteorological stations in the Indian Himalayas.

Daily measurements of standing SD and the total amount of clouds were collected from these meteorological observatories during snow fall season i.e. from October to April in the years 2004–2007. SD is measured in centimetres with a minimum value of 1 cm. Whenever the depth is less than 1 cm, it is considered as snow-free area. Cloud cover observations from the *in situ* stations were recorded with integers ranging from 0 to 9, 0 indicating clear sky and 9 indicating overcast sky. The ground-based cloud observation data and standing SD measurements are used to assess the accuracy of the MODIS daily snow products.

MODIS snow cover product

On 18 December 1999, Moderate Resolution Imaging Spectroradiometer (MODIS) sensor was launched on the Terra spacecraft. The purpose of MODIS is to measure surface temperature, ocean colour, global vegetation and deforestation, cloud and aerosols, and snow cover. The first Earth view data acquisition was made on 24 February 2000. The MODIS snow cover products are one of the many geophysical products derived from MODIS data. The relatively higher spatial resolution (500 m), higher spectral resolution (36 bands, range from 0.4 to 14.3 µm), repeatability (Daily), snow/cloud discrimination capability, automated snow mapping algorithm and global coverage have tremendously improved the potential of MODIS snow cover maps. The MODIS snow product collection is composed of products covering a range of spatial and temporal resolutions, from 500 m to 0.05°, and from swath to daily, to 8-day, to monthly. Snow data products are produced as a series of seven products. The sequence begins as a swath (scene) at a nominal pixel spatial resolution of 500 m with nominal swath coverage of 2330 km (across track) by 2030 km (along track, five minutes of MODIS scans). MODIS snow cover products are continuously updating from MOD10_L2 level 2 products to MODIS10A1 level, 3, 4 and 5 products. In addition to these, MOD10_CG and

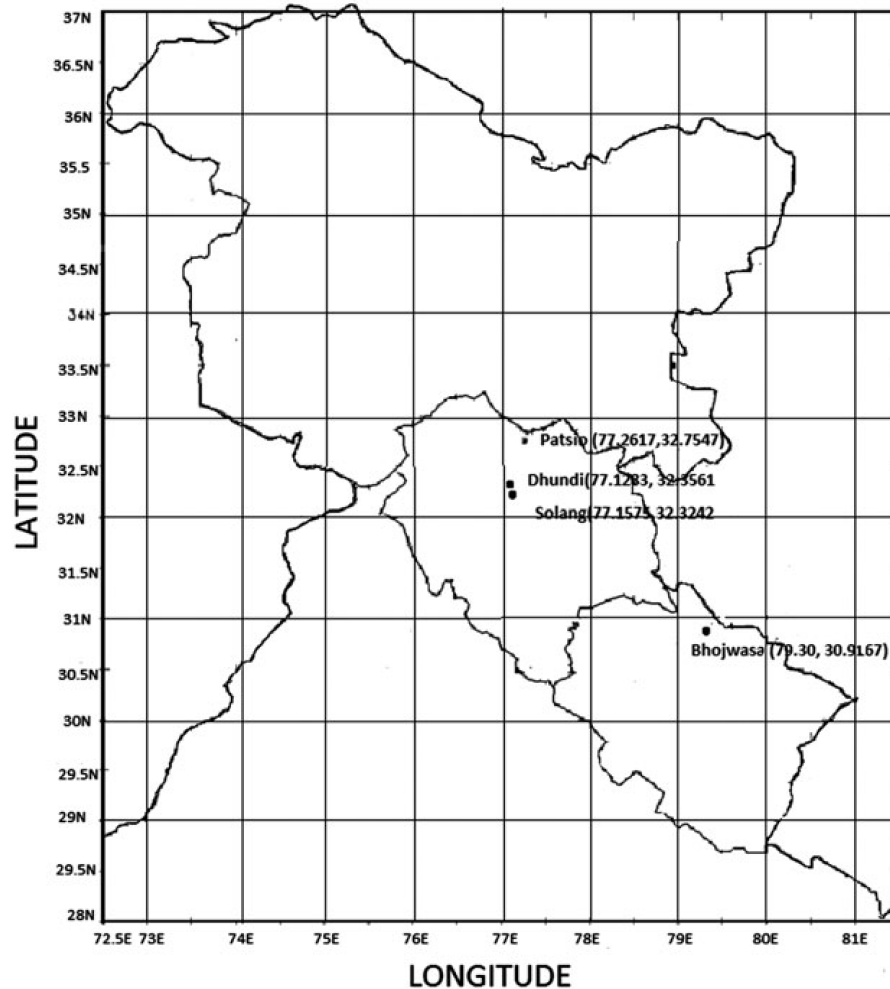


Figure 1. Spatial distribution of meteorological stations in the Indian Himalayas.

MOD10_8D are created for the 'Climate Modelling Grid (CMG)' at 0.05° resolution, with both daily and 8-day composite product, and can be used for global-scale climate models (GCM). Each snow product in the sequence after the swath product assimilates accuracy and error from the preceding product (Riggs et al. 2006). A summarized listing of the sequence of products is given in Table 1.

MOD10A1 snow cover maps

The MODIS/Terra Snow Cover Daily L3 Global 500 m Grid (MOD10A1) contains snow cover, snow albedo, fractional snow cover, and quality assessment (QA) data in compressed Hierarchical Data Format-Earth Observing System (HDF-EOS) format along with corresponding metadata. The MOD10A1 daily product is gridded in equal area tiles. Each tile consists of a 1200 km by 1200 km data array, which corresponds to 2400 by 2400 pixels at 500 m resolution in a sinusoidal projection.

The MODIS snow cover detection algorithm

The MOD10A1 daily snow cover product is the result of selecting an observation from the multiple observations of the day mapped to each cell of the MOD10_L2G swath

Table 1. Summary of the MODIS snow data products.

Earth Science data type (ESDT)	Product Level	Nominal data array dimensions	Spatial resolution	Temporal resolution	Map projection
MOD10_L2	L2	1354 km by 2000 km	500 m	Swath (scene)	None, (lat, lonReferenced)
MOD10L2G	L2G	1200 km by 1200 km	500 m	Day of multiple coincident swaths	Sinusoidal
MOD10A1	L3	1200 km by 1200 km	500 m	Day	Sinusoidal
MOD10A2	L3	1200 km by 1200 km	500 m	Eight days	Sinusoidal
MOD10C1	L3	360° by 180° (Global)	0.05° by 0.05°	Day	Geographic
MOD10C2	L3	360° by 180° (Global)	0.05° by 0.05°	Eight day	Geographic
MOD10CM	L3	360° by 180° (Global)	0.05° by 0.05°	Monthly	Geographic

Source: Riggs et al. 2006 MODIS Snow Products User Guide to Collection 5.

product using a scoring algorithm (Hall et al. 2002). In principle, the method is to select the observation nearest to the nadir with the greatest coverage at the highest solar elevation angle that is mapped onto the grid cell so that the number of pixels with a large off-nadir look angle is minimized. (Xiaodong et al. 2011).

Snow has high reflectance in the visible band (MODIS band 4) and low reflectance in the short-wave infrared band (MODIS band 6). The combination of these two properties allows the detection of snow (Hall et al. 1995) The MODIS snow product classification is based on the normalized difference snow index (NDSI), defined as:

$$\text{NDSI} = \frac{\text{BAND4} - \text{BAND6}}{\text{BAND4} + \text{BAND6}}$$

NDSI is utilized in the fully automated algorithm through a series of thresholds (Hall et al. 2002). A pixel in a non-densely forested region is mapped as snow if the NDSI value greater than or equal to 0.4. In case of a densely forested area NDVI (Normalized difference vegetation Index) values which are calculated based on band 1 and band 2, are used in addition to the NDSI. Pixels with an NDSI of greater than 0.4 and the pixels with slightly-lower NDSI, but high NDVI values (greater than 0.1, (Klein et al. 1998)) can be considered as snow-covered forests. As snow and water both have high NDSI values, an absolute reflectance in MODIS band 2 (841–876 nm) of greater than 11%, is used to separate the snow from water. Other criterion for snow is that reflectance in MODIS band 4 must be greater or equal to 10%, which helps prevent dark targets from being classified as snow despite high NDSI values (Klein & Barnett 2003). A thermal mask using infrared bands 31 (10.78–11.28 μm) and 32 (11.77–12.27 μm) is used to further differentiate snow from clouds and to remove pixels that may be erroneously mapped as snow such as desert sand (Hall et al. 2002; Salomonson & Appel 2004; Simic et al. 2004). If the temperature of a pixel is >283K then the pixel will not be mapped as snow (Riggs et al. 2006).

The snow algorithm identifies pixels as being snow, snow-covered lake ice, cloud, land, water or other condition. Results of the snow algorithm are stored as coded integers in the ‘Snow Cover’ Scientific Data Set (SDS) of the HDF-EOS product file. The coded integers are interpreted as listed in Table 2 (Riggs et al. 2006).

The MODIS Snow Products Users’ Guide contains detailed information pertaining to the structure and information contained in all MODIS snow products (Riggs et al. 2006).

Methodology

The MOD10A1 (V005) data have been ordered, and were made available through the Earth Observing System (EOS) data gateway. The MODIS Hierarchical Data Format (HDF) files are converted into img files with the help of ERDAS software. With the help of same software, analysis of the snow cover condition at the user-defined locations (i.e. Bhojwasa, Dhundi, Solang and Patsio) was performed. The coded pixel values were extracted from the image using ERDAS software and the value attribute table was prepared. Then using the Excel software further analysis and comparative studies between MODIS and observatory *in situ* data have been carried out for all observatory locations.

Table 2. Interpretation key for MODIS snow product.

Coded integer value	Code meaning
255	Fill Data – no data expected for pixel
254	Saturated MODIS sensor detector
253	Dead MODIS sensor detector
200	Snow
100	Snow-Covered Lake Ice
50	Cloud Obscured
39	Openwater (Ocean)
37	Lake or Inland Water
25	Land – no snow detected i.e. Snow-free land
11	Darkness (Night), terminator or polar
1	No Decision
0	Sensor Data Missing

All these observations from the *in situ* stations were recorded two times per day, one in the morning and the other in the afternoon. Since the Terra satellite passes locally at 10 am to 12 noon, the average observatory values were considered. It should be noted that the time discrepancy in the observations could produce some uncertainty in the comparison, but it is the best available data for the study.

Result and analysis

MODIS MOD10A1 image products collected from NSIDC have been analysed using ERDAS software and are broadly classified as snow, cloud and snow-free (bare land). Summary of these data for the entire duration from 2004 to 2007 is presented in column charts as shown in Figure 2.

All the images have been collected for the months of October to April, as snowfall happens only during these months. From the observations, it has been found that snow cover duration is more at Dhundi and less at Bhojwasa. A substantial amount of data (43.47%) is affected of cloud cover. Cloud cover is a common problem at high altitudes for optical images. Though other sensors like microwave do eliminate the problem of cloud cover, their repeatability is very low. Hence, even with the problem of the cloud

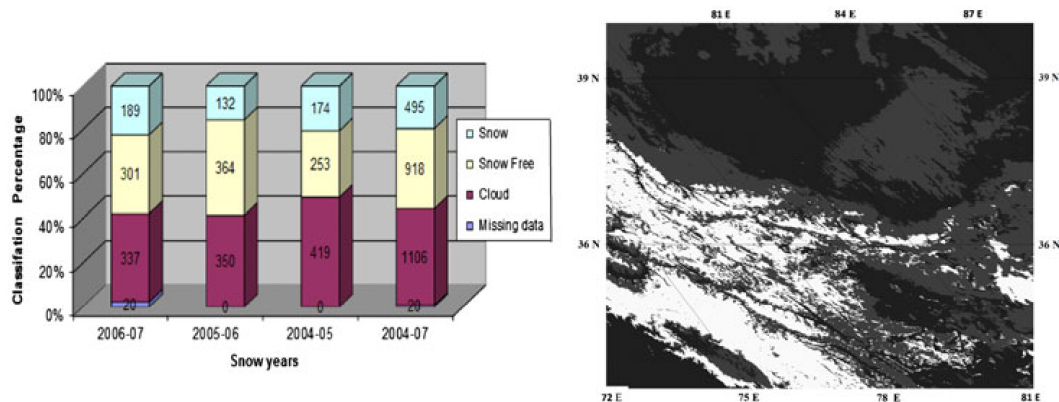


Figure 2. MODIS snow cover classification 2004–2007 with Sample MOD10A1 snow cover daily tile (28 February 2006).

cover, the MODIS data can be considered to have better data availability as compared to the other sensors.

Comparison of MODIS snow maps were made against daily *in situ* SD measurements made at four SASE observatory sites scattered in the Himalayas. These comparisons are not perfect, as a point observation having lesser area around each observatory site is being compared to a large area image obtained by satellite sensing. To understand the overall agreement of the snow cover map, charts have been drawn with daily MODIS classification criteria and *in situ* standing depth for all observatories, year wise. The comparison of MODIS snow cover with SD during 2004–2005, 2005–2006 and 2006–2007 are shown in Figures 3–12 at different observatories. In these charts for the MODIS data, the rectangular points against 25 represent snow-free, against 50 represent cloud affected, against 200 represent snow cover and 0 represents missing data. Whenever the standing snow is nil, its corresponding rectangular point on the MODIS data should be 25, which is correct. If the same appears as 50, there is cloud cover due to which MODIS is not able to recognize the actual condition. However, if the same appears as 200 then the MODIS has wrongly classified snow-free as snow. Similarly, if standing snow is present, its corresponding rectangular point on the MODIS data should be 200 or 50 but not 25.

From the Figures 3–12, it is observed that MODIS fails to map whenever the SD is less and succeeds when SD is more. In Figure 13, the bar graph is drawn to understand the relation between SD and MODIS MOD10A1 product accuracy for different SD classes. SD is classified into four classes based on measured standing SD at the observatories as follows:

- (1) Class I: SD from 1 cm up to 10 cm
- (2) Class II: SD Greater than 10 cm up to 20 cm
- (3) Class III: SD Greater than 20 cm up to 30 cm
- (4) Class IV: SD Greater than 30 cm

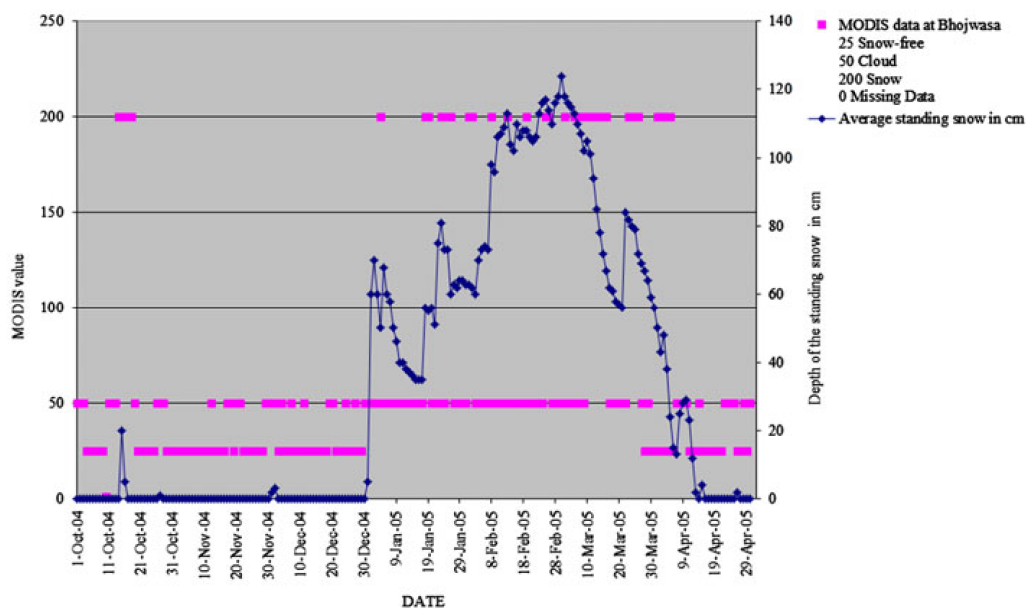


Figure 3. Comparison of MODIS snow cover with snow depth at Bhojwasa 2004–2005.

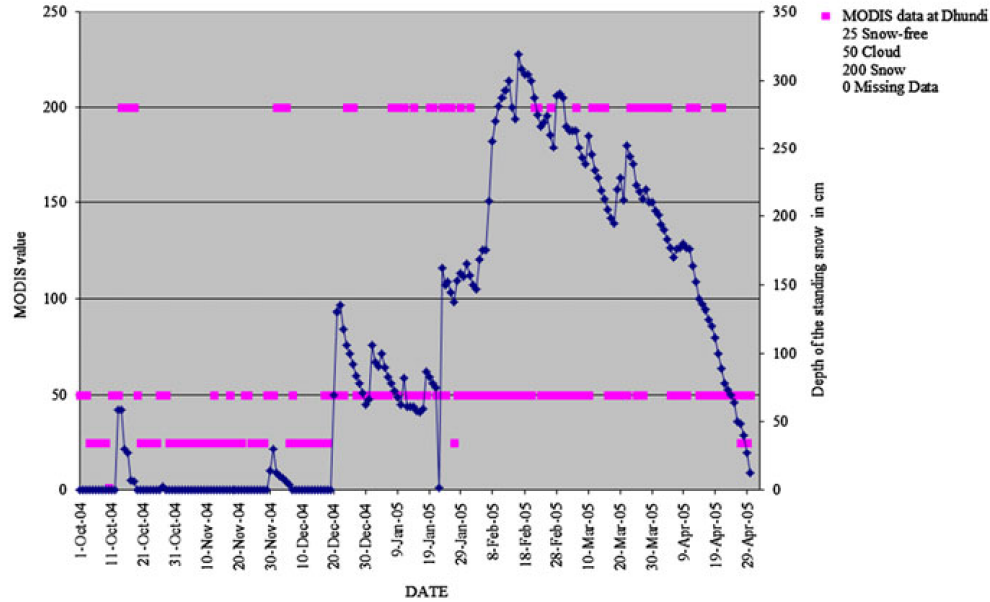


Figure 4. Comparison of MODIS snow cover with snow depth at Dhundi 2004–2005.

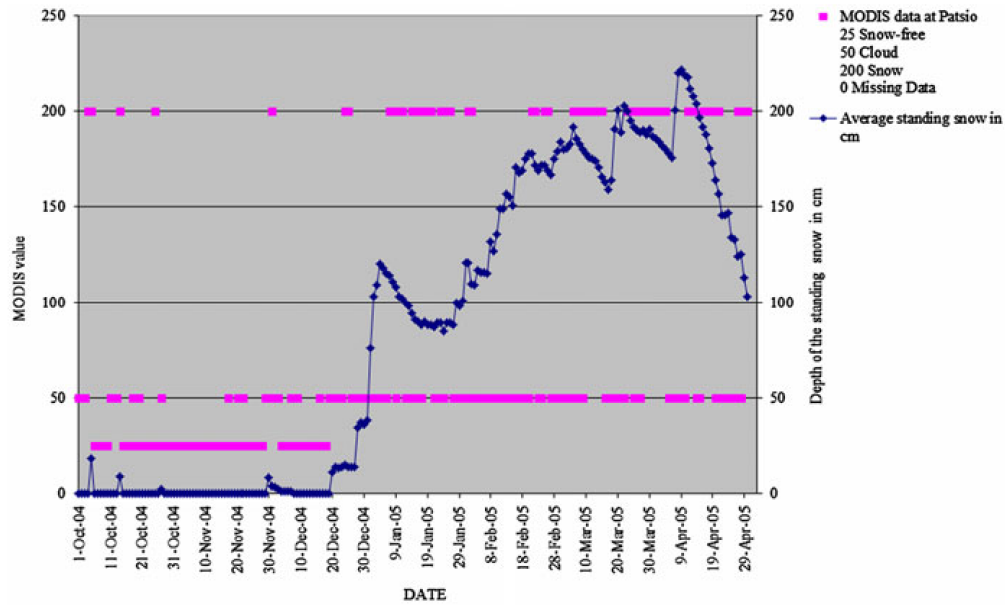


Figure 5. Comparison of MODIS snow cover with snow depth at Patsio 2004–2005.

From the Figure 13, for all the observatories in class IV, accuracy is greater than 84.62%, whereas in class I accuracy is comparatively very less i.e. less than 50%. Minimum percentage of accuracy is observed as 13.25% at Bhojwasa in Class I whereas maximum percentage accuracy is observed at Patsio in class IV. The average accuracy is observed to be 30.28% in class I, 55.24% in class II, 72% in class III and 92.63% in class IV. It clearly shows that the MODIS accuracy is dependent on the SD and it increases as the SD increases.

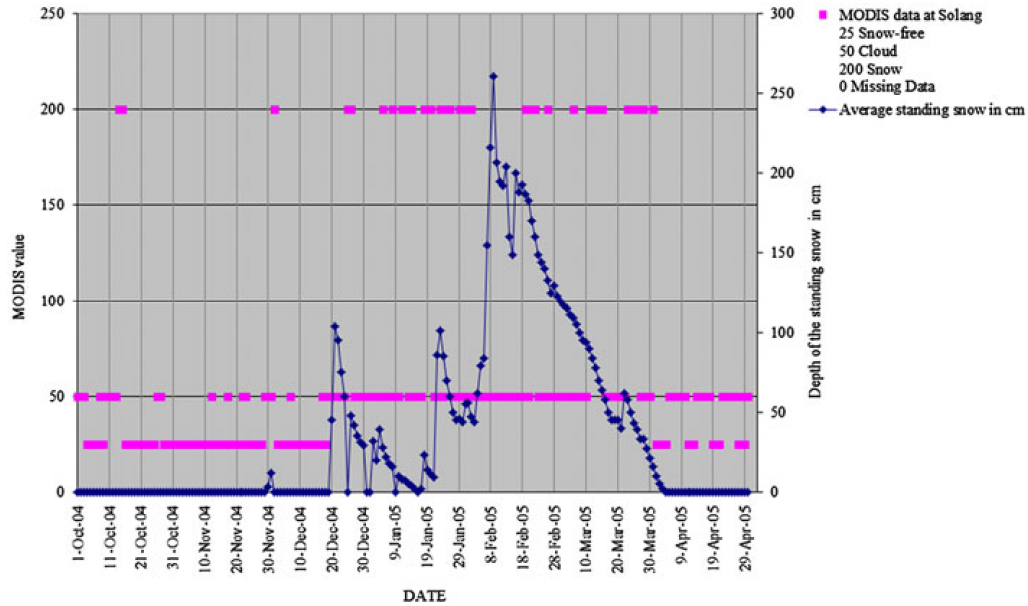


Figure 6. Comparison of MODIS snow cover with snow depth at Solang 2004–2005.

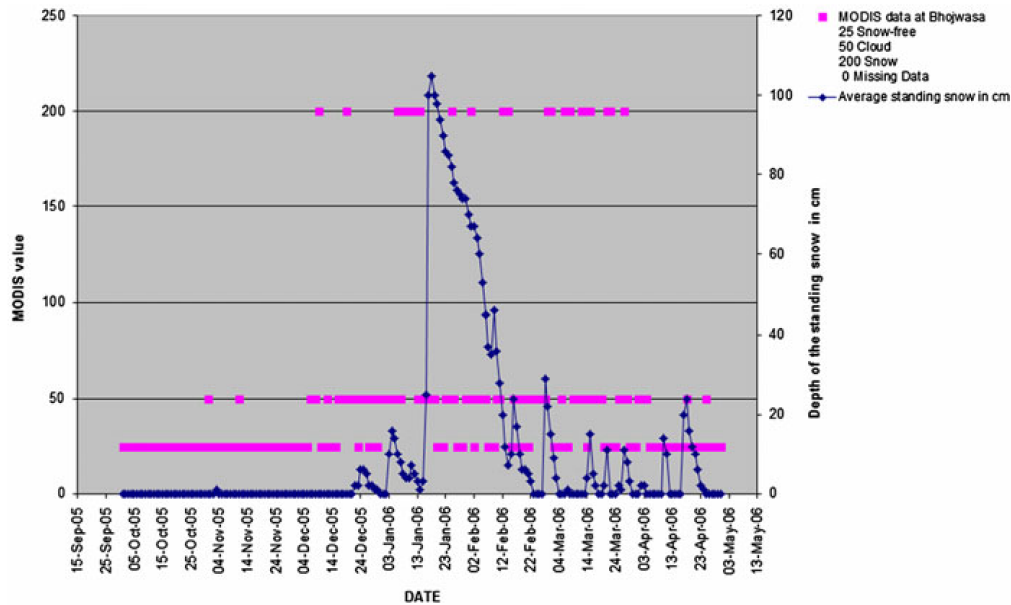


Figure 7. Comparison of MODIS snow cover with snow depth at Bhojwasa 2005–2006.

To quantify numerically, a confusion matrix has been prepared. A confusion matrix or error matrix is a widely used technique to assess the accuracy of remotely sensed data through comparisons with reference data.

MODIS data are classified into an extra category of cloud cover in addition to the two categories of snow and snow-free (bare land) which are common with ground observatory data. In order to take the effect of cloud cover in the statistical analysis, the data of the day prior to cloud cover are assumed to be the data for the cloud cover.

With this assumption, the MODIS data are mapped into two classes as in the ground observatory data. The confusion matrices are prepared after taking into

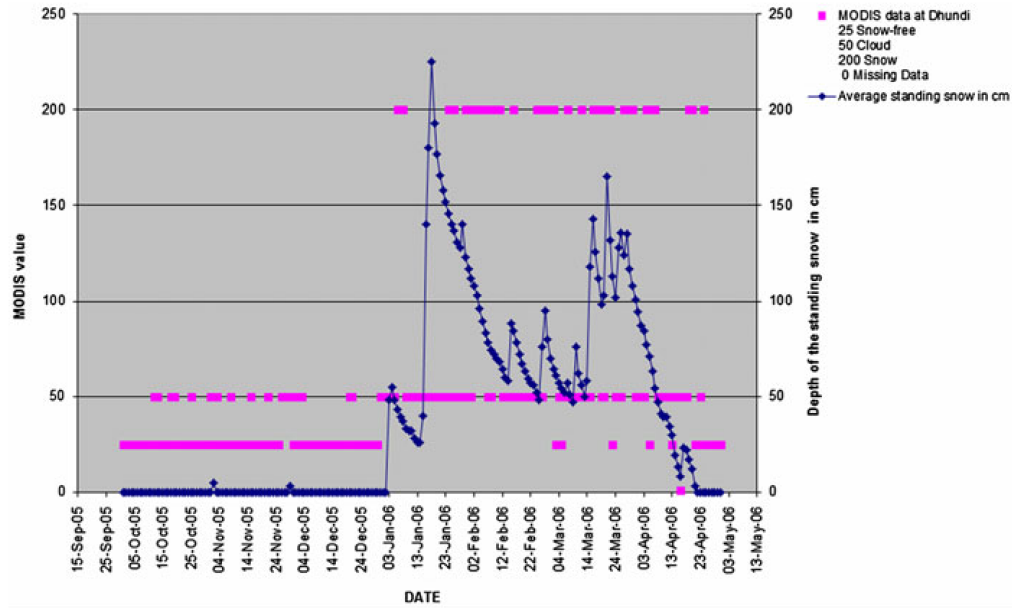


Figure 8. Comparison of MODIS snow cover with snow depth at Dhundi 2005–2006.

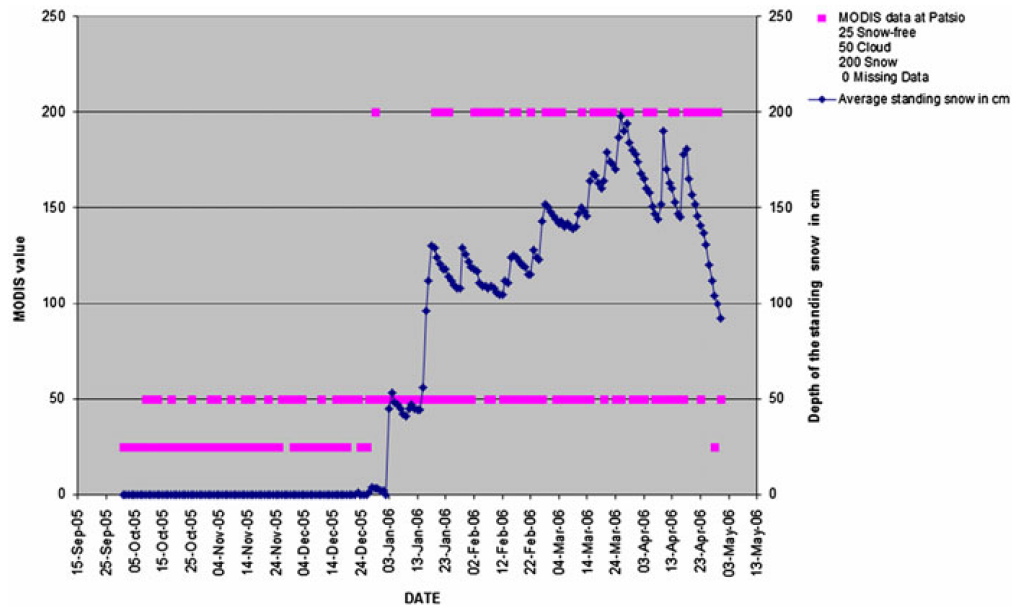


Figure 9. Comparison of MODIS snow cover with snow depth at Patsio 2005–2006.

consideration the above assumption for each observatory year wise as well as for the combined data as shown in Tables 3–6. They illustrate the comparison between all ground observations and MODIS snow cover classification. In each cell, the first number indicates the counts, while the second number shown in the parenthesis indicates the percentage of these counts with respect to the total counts of the same class.

A total of 2120 images have been analysed and classified as 963 (45.42%) having snow and 1157 (54.57%) having bare land (without snow). MOD10A1 product misclassifies 45 (2.1%) of them as snow and 209 (9.85%) of them as bare land. The accuracy

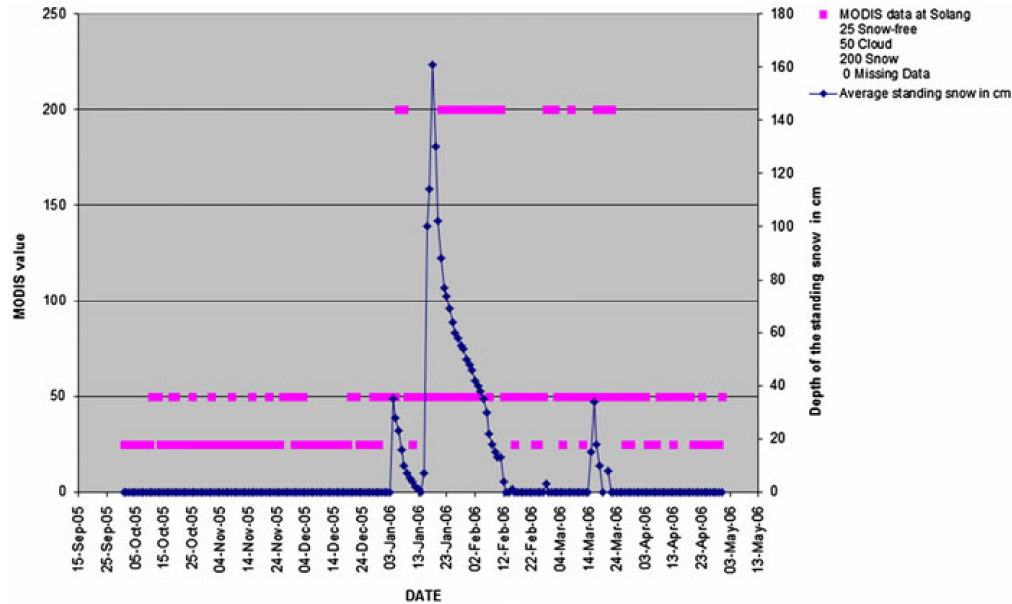


Figure 10. Comparison of MODIS snow cover with snow depth at Solang 2005–2006.

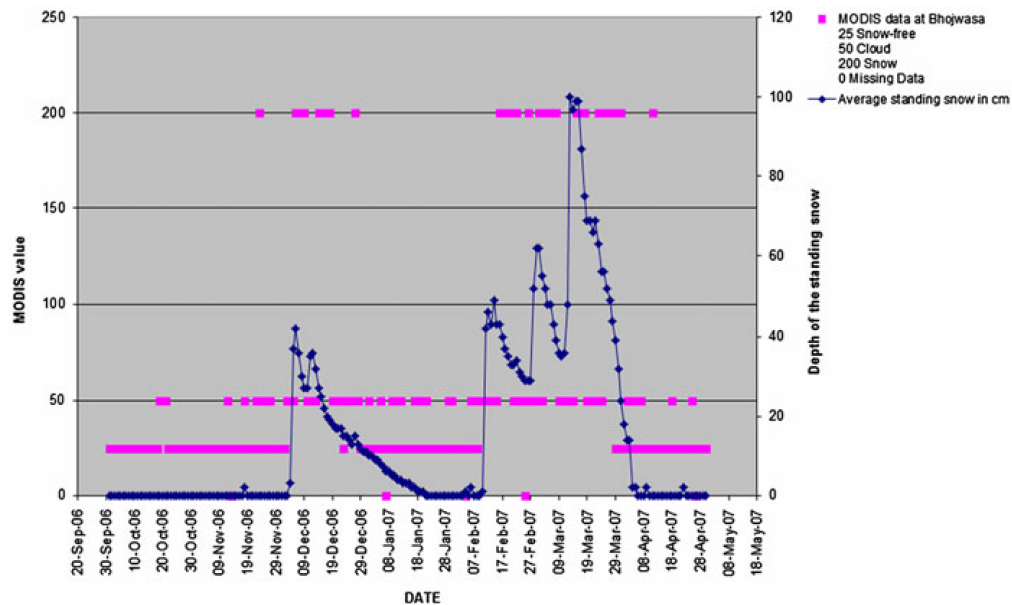


Figure 11. Comparison of MODIS snow cover with snow depth at Bhojwasa 2006–2007.

of bare land classification is 95.47%, where as that of snow is 81.46% and overall agreement of data is 1866 i.e. 88.02%. Overall omission error (missing snow when it present) is 9.85% compared to the commission errors (mapping snow when none is present) of 2.1%. Generally, there is a higher percentage of omission errors as compared to the commission errors

In this study, the Patsio observatory shows the maximum agreement (97.65%) during 2005–2006. Lowest agreement (70.29%) was observed at Bhojwasa in the same year. At the Bhojwasa site, the level of agreement is lesser averaging around 78% as compared to the other three sites which are in the range of 88–91%.

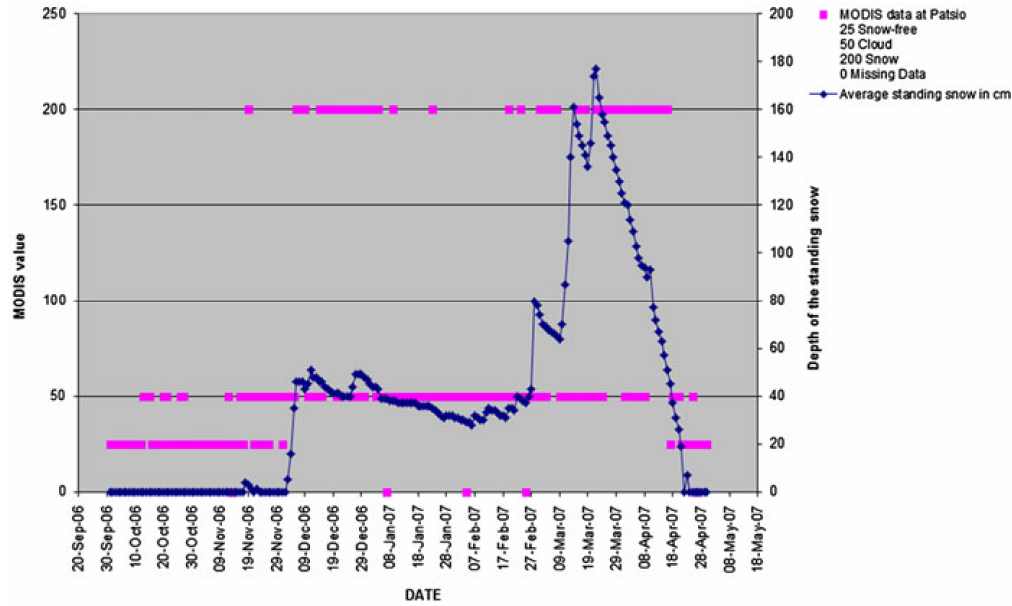


Figure 12. Comparison of MODIS snow cover with snow depth at Patsio 2006–2007.

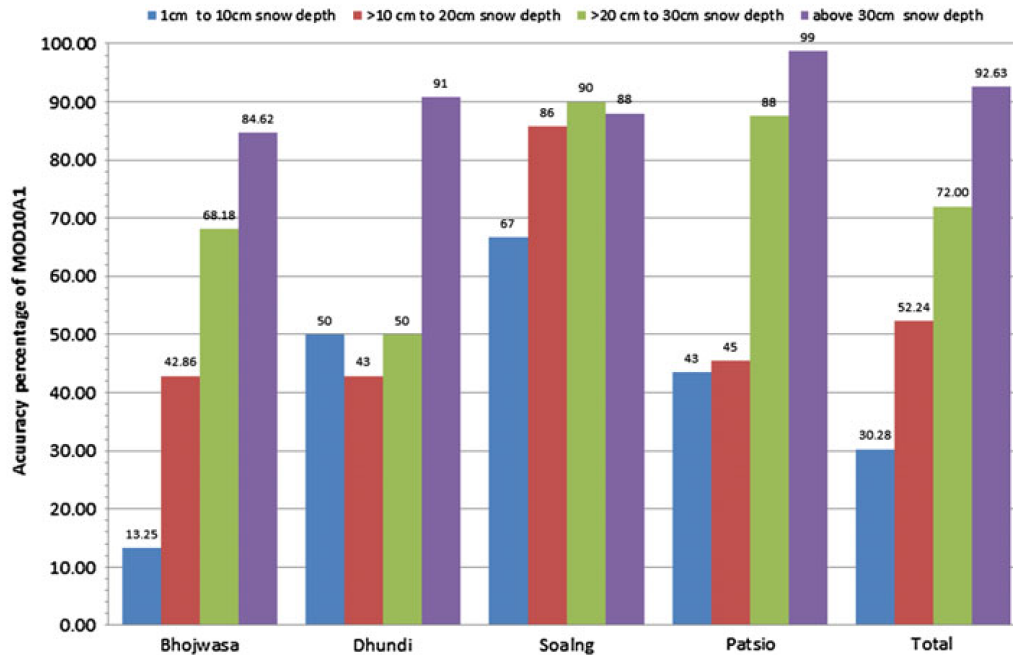


Figure 13. Accuracy of MODIS MOD10A1 snow cover with respect to snow depth class from 2004 to 07.

There is a considerable variation of agreement/disagreement between snow maps and the *in situ* observations from 70.29 to 87.27% made at the Bhojwasa site in comparison with the other sites, Dhundi (89.63–92.46%), Solang (85.38–91.51%) and Patsio (94.34–97.65%).

Reasons for the drastically different snow mapping accuracies are difficult to ascertain. No consistent relationships between site slope, aspect, elevation, relief and snow

Table 3. Confusion matrix comparing observatory and MODIS snow classification 2004–05.

Observatory	2004–05	MODIS			Data agreement
		Snow	Snow-free	Total	
Bhojwasa	Snow	88 (79.28%)	23 (20.73%)	111	185 (87.27%)
	Snow-free	4 (3.97%)	97 (96.04%)	101	
	Total	92	120	212	
Dhundi	Snow	131 (89.73%)	15 (10.28%)	146	196 (92.46%)
	Snow-free	1 (1.52%)	65 (98.49%)	66	
	Total	132	80	212	
Solang	Snow	92 (90.2%)	10 (9.81%)	102	194 (91.51%)
	Snow-free	8 (7.28%)	102 (92.73%)	110	
	Total	100	112	212	
Patsio	Snow	133 (93.01%)	10 (7%)	143	200 (94.34%)
	Snow-free	2 (2.9%)	67 (97.11%)	69	
	Total	135 (135%)	77 (77%)	212	
Total	Snow	403 (87.42%)	58 (12.59%)	461	769 (90.69%)
	Snow-free	21 (5.43%)	366 (94.58%)	387	
	Total	424	424	848	

Table 4. Confusion matrix comparing observatory and MODIS snow classification 2005–06.

Observatory	2005–06	MODIS			Data agreement
		Snow	Snow-free	Total	
Bhojwasa	Snow	35 (38.89%)	55 (61.12%)	90	149 (70.29%)
	Snow-free	8 (6.56%)	114 (93.45%)	122	
	Total	43	169	212	
Dhundi	Snow	90 (81.09%)	21 (18.92%)	111	190 (89.63%)
	Snow-free	1 (1%)	100 (99.01%)	101	
	Total	91	121	212	
Solang	Snow	33 (71.74%)	13 (28.27%)	46	181 (85.38%)
	Snow-free	18 (10.85%)	148 (89.16%)	166	
	Total	51	161	212	
Patsio	Snow	121 (96.8%)	4 (3.2%)	125	207 (97.65%)
	Snow-free	1 (1.15%)	86 (98.86%)	87	
	Total	122 (122%)	90 (90%)	212	
Total	Snow	191 (65.19%)	102 (34.82%)	293	701 (82.67%)
	Snow-free	45 (8.11%)	510 (91.9%)	555	
	Total	236	612	848	

mapping accuracies are readily apparent. Land cover type appears to have the strongest relationship to snow mapping accuracy. Care must be taken in interpreting these results because conditions at individual observatory sites may not be representative of the larger area viewed by satellites, especially in the mountainous terrain. It is possible that snow may be present at elevations above the site, but not at the measuring site itself, which would lead to the high commission errors seen in the MODIS snow maps. Conversely, snow may not be present at levels below the site, but at the site itself, which leads to omission errors. The analysis could be further extended with the help of more observatory data and high-resolution sensor data.

Table 5. Confusion matrix comparing observatory and MODIS snow classification 2006–07.

Observatory	2006–07	MODIS			Data agreement
		Snow	Snow-free	Total	
Bhojwasa	Snow	61 (56.49%)	47 (43.52%)	108	163 (76.89%)
	Snow-free	2 (1.93%)	102 (98.08%)	104	
	Total	63	149	212	
Dhundi	Snow	No Data			—
	Snow-free				
	Total				
Solang	Snow				—
	Snow-free				
	Total				
Patsio	Snow	134 (92.42%)	11 (7.59%)	145	201 (94.82%)
	Snow-free	0 (0%)	67 (100%)	67	
	Total	134	78	212	
Total	Snow	195 (77.08%)	58 (22.93%)	253	364 (85.85%)
	Snow-free	2 (1.17%)	169 (98.84%)	171	
	Total	197	227	424	

Table 6. Confusion matrix comparing observatory and MODIS snow classification 2004–07.

Observatory	2004–07	MODIS			Data agreement
		Snow	Snow-free	Total	
Bhojwasa (2004–07)	Snow	184(59.55%)	125 (40.45%)	309	497 (78.15%)
	Snow-free	14 (4.29%)	313 (95.72%)	327	
	Total	198	438	636	
Dundi (2004–06)	Snow	221 (86%)	36 (14.01%)	257	386 (91.04%)
	Snow-free	2 (1.2%)	165 (98.81%)	167	
	Total	223	201	424	
Solang (2004–06)	Snow	125(84.46%)	23 (15.54%)	148	375 (88.45%)
	Snow-free	26 (9.43%)	250 (90.58%)	276	
	Total	151	273	424	
Patsio (2004–07)	Snow	388(93.95)	25 (6.05%)	413	608 (95.60%)
	Snow-free	03(1.34%)	220 (98.66%)	223	
	Total	391	245	636	
Total	Snow	918(81.46%)	209 (18.54%)	1127	1866 (88.02%)
	Snow-free	45 (4.53%)	948 (95.47%)	993	
	Total	963	1157	2120	

Conclusions

The MOD10A1 product has been validated for the Himalayan territory and shows a sufficiently high degree of agreement with the limited available *in situ* data obtained from the SASE observatories. The accuracy of the product is relatively less when SD is less than 10 cm.

Due to accessibility limitations, it is difficult, though not impossible, to estimate the total snow accumulation, snow cover extent and snow depletion curve for the entire region based on only ground observations. The MODIS snow cover maps, however, may be utilized in developing snow depletion curves which in turn can be used as an input for the snowmelt runoff model.

Clouds are a major reason for degradation in the accuracy of MODIS snow classification. The MOD10A2 product may have higher classification accuracy for both snow and land than MOD10A1, as the MODIS-MOD10A2 8-day product minimizes cloud cover and maximizes snow cover. Thus, the MOD10A2 product may be selected for further study.

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